



5441 - A JWST Sample of Extragalactic Red Supergiants

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

NIRCam and MIRI imaging from JWST early release and Cycles 1 and 2 programs provides a vast trove of resolved red supergiant populations in nearby galaxies, increasing by more than a factor of 2 the number of these stars that can be studied in detail from the optical to mid-infrared. This large population of red supergiants can be used to look for rare forms of extreme mass loss predicted for highly-evolved stars, measure the complete bolometric luminosity of red supergiants out to 10 microns, and analyze the dependence of red supergiant properties on metallicity and local environment, especially the uncertain maximum luminosity of red supergiants (the Humphreys-Davidson limit). We propose an archival research program to analyze NIRCam and MIRI imaging covering 11 galaxies within 25 Mpc and totaling 155 unique data sets, identify an estimated 20,000 red supergiants in these data, compare their wide- and medium-band photometry to model red supergiant spectral energy distributions, and analyze the distribution of red supergiant luminosities, spectral types, and mass-loss rates. The final product of this analysis will be the largest population to date of photometrically-identified red supergiants as well as their photometry and basic physical properties. Our program will provide lasting legacy value for JWST and a framework for future studies of resolved massive star populations in NIRCam and MIRI data.

OBSERVING DESCRIPTION

This archival research proposal aims to analyze 155 NIRCam and MIRI data sets covering 12 galaxies within 25 Mpc in order to find red supergiants in their stellar populations. These data comprise all available medium- and wide-band NIRCam and MIRI filters within 1 arcminute of the target galaxies, totaling anywhere from 6-29 filters per galaxy but always with a single "optical" (<1.5 micron; either F090W or F115W), "near-infrared" (F150W), and "mid-infrared" (F444W) filter in order to optimally cover red supergiant spectral energy distributions. We will analyze all such data, use them to isolate red supergiants in color-magnitude space, and analyze the distributions of these stars as a function of their basic physical properties and local environment in order to achieve our overall science goals.